Crop technology





Huge amounts of wheat are grown in Australia. Crops grown close together can suffer from outbreaks of disease. Scientists have identified a **gene** that makes a protein called **Lr34 transporter protein**. This protein protects wheat plants against three diseases – leaf rust, stripe rust and powdery mildew – that can severely damage wheat crops. Now that the gene sequence of the protein has been identified, it can be implanted in wheat plants and bred widely.

☞ **1** Use the word bank to complete the following passage.

Word bank

growing benefit protects living cooperating
future passes Millions disease gene

Plants are _____ things. They can get sick. _____ of plants grow together in a farming area such as a wheat- _____ area. If one plant is sick and _____ the disease on, a whole year's crop can be wiped out. If a few plants do not get sick, this may mean they have a gene that _____ them.

If a whole crop is planted with that protecting _____, the crop is safe from the _____. The Lr34 gene will protect _____ crops against three fungus infections. Earlier attempts to prevent disease have had short term results, but the _____ of Lr34 will last for a long time.

Australia's CSIRO scientists are _____ with others around the world to improve crops. A new type of rice plant is being developed which will produce much more food, for example.

2 Write a sentence explaining how the problem of world hunger could be partly solved by the new rice plant that is being developed. _____

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